

SOPHIA MENGJIA LI (MSc)

Machine Learning Engineer / AI Researcher

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[LinkedIn](#) • [GitHub](#) • [Website](#) • [Google Scholar](#)

EDUCATION

Master of Science in Applied Computing (MScAC) | University of Toronto September 2026 – December 2027
Department of Computer Science, Faculty of Medicine • Focus in Machine Learning Engineering
Courses: Neural Networks and Deep Learning, Computational Imaging and 3D Sensing, Automated Reasoning with Machine Learning

Honours Bachelor of Science (HBSc) | University of Toronto September 2022 – June 2026
Department of Computer Science • Major in Computer Science, specialist in Bioinformatics & Computational Biology
GPA: 3.9/4.0 • Dean's List Scholar (2022 – 2026) • B. Arthur Bensley Scholarship in Life Sciences (2025)
Courses: Machine Learning, Artificial Intelligence, Data Structures and Algorithms, Probability and Statistics, Linear Algebra, Calculus

RELEVANT EXPERIENCE

Machine Learning Researcher | Princess Margaret Cancer Centre, UHN September 2025 – Present
Computational Cancer Genomics Lab, Dr. Sushant Kumar: Automated Segmentation of IHC and H&E Staining

- Designed a multi-modal cascaded diffusion model integrating four structured-data foundation models to synthesize high-resolution images from tabular multi-omics inputs, **achieving an 83% match in morphological diversity** across image classes using PyTorch
- Led development of a 24.4M-parameter U-Net segmentation pipeline and image-analysis toolkit extensions, trained on 10M image patches for automated structure segmentation and multi-class object quantification, **achieving an expert-validated 92% Dice score**

Full-Stack Software Engineer | Stealth E-Commerce Platform Startup September 2025 – May 2026
Designed backend database system for deployed e-commerce marketplace serving over 100k university students, optimizing API fetching with bucket updates resulting in a **65% decrease in page loading times** using AWS cloud servers, supabase, and PostgreSQL

Computer Vision Research Intern | Lunenfeld-Tanenbaum Research Institute, Mount Sinai May 2025 – February 2026
Campbell Lab, Dr. Kieran Campbell: Computer Vision Methods for Highly Multiplexed Biological Images
Built an end-to-end image analysis toolkit (ResNet-34 encoder convolutional autoencoder) optimized for highly multiplexed, high-dimensional imaging data, **reducing runtime by 30% over existing non-optimized toolkits** using Python and PyTorch; cited at *Nature*
Designed a novel stratified, morphology-specific information saturation sampling protocol for benchmarking imaging modalities against ground truth, capturing the spatial heterogeneity in complex tissue-like environments using PyTorch Lightning; cited at *Nature*

Data Science Research Intern | Massachusetts General Hospital, Harvard Medical School September 2024 – July 2025
Sadreyev Lab, Dr. Ruslan I Sadreyev: Molecular Mechanisms of Acute Myeloid Leukemia
Engineered multi-modal sequencing data workflows integrating the largest cross-cohort datasets assembled for the task, applying unsupervised clustering (fuzzy K-means, K=3 and K=5) for integrative differential subtype analysis using R

Lead Systems Software Engineer | Victoria University, University of Toronto May 2024 – September 2026
Led end-to-end architectural and data retention protocol redesign of a university-scale intranet platform serving over 10,000 users
Normalized database schemas resulting in a **17% reduction in storage redundancy** using PostgreSQL and MongoDB

SELECTED PUBLICATIONS

- Nowlan F, et al. (incl. Li SM), "Integrated spatial proteomics of human PDAC uncover an expanded tumour-immune-stroma spectrum with genomic associations," *Nature*, 2026

SELECTED RESEARCH OUTPUTS

- Image Segmentation Pipeline** (Python) – Tuned U-Net for image segmentation and multi-class object quantification/classification
- Image Analysis Toolkit** (Python) – Toolkit for image mask generation, patch extraction, and preprocessing of highly multiplexed images

TECHNICAL SKILLS

ML/DL: Fluent in PyTorch Lightning, TensorFlow, scikit-learn • Experienced with Machine Learning, Computer Vision, Deep Learning
Languages: Fluent in Python, R, C/C++, Java, and PostgreSQL • Experienced in MySQL, TypeScript, JavaScript, Bash, Assembly
Infrastructure: Experienced in AWS Cloud Computing Services, Vercel, Supabase